

Innovation under Uncertainty: Perspectives from Associative Learning ^{*}

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Abstract

I propose an associative-learning framework to study how firms innovate when they are uncertain about the data generating process of growth opportunities in technologies. Firms hold Gaussian-process beliefs about the payoffs over technological space and update them using market feedback to their patents. Because feedback in one field informs the firm about the values of technologically related fields, learning generates conditional path dependence: even holding accumulated technological knowledge fixed, different signal histories lead firms to pursue different innovation paths. Using patent-level market responses and citation-based measures of technological proximity for U.S. public firms, I document that firms direct subsequent patenting toward fields associated with more favorable signals, are more likely to enter new fields near higher-signal experience, and avoid entering the technological neighborhoods of fields with unfavorable signals. Embedding the learning mechanism in a quantitative endogenous-innovation environment calibrated to U.S. patent data, I show that field-specific market optimism can generate a persistent reallocation of innovation toward high-sentiment technologies. Most of this response comes from firms intensifying innovation in fields where they are already active, rather than from encouraging other firms entering those fields.

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1 Introduction

The modern growth literature allows innovation outcomes to be stochastic but generally assumes that firms know the distribution of returns to innovative activity. However, in reality, the innovative activities are highly uncertain, and the innovators often hold miscalibrated beliefs about how valuable their innovation is before it is released to the market. They discover the value of innovation by learning from the market responses to the innovation. A striking recent example is the reception of ChatGPT. Before its public release in late 2022, the product was not widely viewed within OpenAI as a transformative innovation, and its developers reportedly expected only modest adoption.¹ The extraordinary response from users following its release was far beyond the expectations of the developers. The company subsequently redirected its R&D resources towards GPT-based products from 2023 onwards.²

The motivating example brings up an important question: How do uncertainty and learning shape firms’ allocation of R&D resources and direction of innovation? To answer the question, I embed associative learning in a canonical endogenous-growth environment, in which firms are uncertain about effective innovation step sizes across technological fields and learn about them from the market responses generated by their patents. The learning model predicts an *history-dependent* path of endogenous innovation, which is empirically supported by the new facts I document from the patenting data of U.S. public firms. The learning model also makes it possible for non-fundamental factors, such as sentiments and narratives, to affect the direction of technological growth. I explore this possibility using a calibrated model.

In the model, firms own patents in multiple technological fields, and grow through filing new patents in the fields that they already own (which I refer to as “internal innovation”) and those which they have not owned yet (which I refer to as “external innovation”).³ The innovation decisions are subject to imperfect knowledge about the value of patents in the technological fields. Specifically, firms do not know how effective step sizes vary across technological fields, which needs to be learned from market signals following successful patents. The firms’ subjective uncertainty and learning about patent values over the technological

¹For instance, in a podcast interview, Jiayi Weng, a key developer in the ChatGPT team, said “We developed the chatbot to elicit some real-world user data... we expect it to have 10k or 20k active users and close it if it drops to 0... but the market response was totally beyond expectation: the user base grew exponentially...”. For details, see <https://music.youtube.com/podcast/I0DrCsDf30s>.

²There was a massive media coverage on this. For example, <https://www.technologyreview.com/2023/03/03/1069311/inside-story-oral-history-how-chatgpt-built-openai/>.

³This is a standard setup in the canonical quality-ladder literature, e.g. [Akcigit and Kerr \(2018\)](#).

space are modeled by a Gaussian process, which has two features. First, it is *Bayesian*: firms update their beliefs about the unknown patenting values in a given field by Kalman filtering with the noisy market signals the patents generate. Moreover, it is *associative*: learning about the value of patents in one technology may change firms’ beliefs about the value of patenting in similar technologies more than in less relevant technologies. For example, a signal generated by a patent in signal modulation (CPC subclass H03C) is more informative about innovation prospects in signal demodulation (CPC subclass H03D) than in a technologically distant chemistry field such as heterocyclic compounds (CPC subclass C07D). This cross-field learning is handled by the kernel function (which gives the covariance between the values of different technologies) specified in the Gaussian process.

The model generates rich predictions on the innovation patterns of the firm. First, associative learning makes firms’ choice of innovation history-dependent: firms with different histories of signal realizations generally choose different levels of innovation intensity (the probability of having a successful patent), internal or external, even if they are of the same size (by number of technological fields they filed patents in) or have filed same number of patents in the field, and firms tend to choose higher innovation intensity on the fields with high realizations of signals. This prediction makes the associative learning model observationally different from the canonical quality-ladder model (e.g. Klette and Kortum (2004), Lentz and Mortensen (2008), Akcigit and Kerr (2018)), where the firms of the same size choose the same level of innovation intensity, and the directed innovation model (e.g. Acemoglu (2002), Acemoglu et al. (2012)), where path dependence is driven by the existing stock of knowledge in the field.

Additionally, the associative learning model generates a unique geometric pattern of external search in the technological space: firms are more likely to enter a new technological field near their own fields that experienced high realizations of signals, and avoid external fields close to their low-signal own fields. Intuitively, under associative learning, as a firm receives positive signal on one field, it will update its belief about the values of innovating in technologically nearby fields. Hence, the firm is more likely to pick these fields as the target of external search and spend more R&D resources on them, which generates a small step size in technological space. Similarly, if the firm instead receives negative signals on an internal field, its subjective evaluation on the nearby fields will be lowered, which leads the firm to take larger steps and move farther from these fields in technological space.

To validate the model, I test the predictions using the patenting history of U.S. public firms from 1976 to 2020. I combine the universe of patents filed or granted by U.S. firms

from USPTO with the patent-firm linkage in [Kogan, Papanikolaou, Seru, and Stoffman \(2017\)](#), which also contains a measure of patent value constructed from abnormal stock price movement in a three-day window around the grant date of each patent by U.S. public firms. I use the component of this patent value that is not explained by its citations, the patenting firm, or contemporaneous field conditions as a proxy for the market signal. I define the technological space as CPC subclasses from USPTO, and use the backward citation similarity between CPC subclasses to construct a proxy of distance between fields, following the classical approach by [Jaffe \(1986\)](#) and [Bloom, Schankerman, and Van Reenen \(2013\)](#).

I document the following facts from the combined dataset. First, within a firm and year, higher past signals predict greater subsequent innovation in the corresponding existing fields, after controlling for the firm size and knowledge stock in the field. A one-unit increase in average realized signals increases the probability of subsequent innovation in the field by 12.5%. Second, new-field entry is systematically directed toward technological regions receiving stronger signals: firms are more likely to enter a candidate external field that is closer to the fields that have accumulated high realizations of signals and are technologically similar to the candidate field. Within a firm-year, a candidate field at the 90th percentile of the similarity-weighted signal is 7.7% more likely to be entered than one at the 10th percentile. Firms move away from fields that are technologically similar to those associated with unfavorable feedback. Lastly, the distance between a firm’s new field and its existing field in the technological space is negatively correlated with the signal realizations of the existing field. A one-within-firm-year-standard-deviation increase in the signal implies a reduction in technological distance from a new field by 2.01 percentile points closer. These empirical patterns are consistent with the predictions of the associative learning model.

The past four decades of U.S. stock market witnessed ebbs and flows of sentiments in various technological fields, from the Internet and ICT revolution in 1990s to the recent AI boom. The associative learning model makes it possible for market sentiments of this kind to shape the path of innovation. In the last part of the paper, I calibrate the associative-learning growth model and use it to study the response of firms’ innovation behaviors to a wave of optimism by 2 standard deviations of signal noise for a 10-year episode. An optimistic signal shift applied for ten years to a connected block of technologies raises the block’s patent share from 7.76 percent in the paired benchmark to 31.29 percent at the end of the episode. The reallocation persists after sentiment disappears: the difference remains 6.77 percentage points ten years later and 1.08 percentage points after twenty years. The strong and persistent effects of sentiment shock come from a novel feedback loop between patents and learning: as firms in the shocked fields file patents in the sentiment episode, the optimism

wave shifts the signal realization upwards, which makes firms increase their evaluations of patenting values in these fields. They allocate more resources to this field, which results in a higher probability of success and generates more favorable signals in the field, further reinforcing their beliefs.

When the same optimistic shift applies to all fields, cumulative patenting rises by 73.1 percent during the episode relative to the no-sentiment benchmark, and the effect remains persistent. Interestingly, the boom is driven primarily by internal innovation, which rises by 164.5 percent cumulatively during the episode, compared with 26.0 percent for external innovation. The share of external patents keeps declining during the sentiment episode. This is because the feedback loop between patent and learning is weaker for external innovation: firms update their beliefs about the external fields by extrapolating from the higher beliefs in their internal fields, which is not repeatedly reinforced by the signals generated from that external field. The learning model offers one possible interpretation of three salient patterns in the U.S. economy in the past four decades: persistently booming U.S. stock market return, persistently increasing number of patents in aggregate, and persistently declining share of external patents in the economy.

Related Literature First, this paper contributes to the discussions of heterogeneity of innovation in the endogenous growth literature. The state-of-the-art quality-ladder literature (Grossman and Helpman (1991), Aghion and Howitt (1992), Klette and Kortum (2004), Lentz and Mortensen (2008), Akcigit and Kerr (2018), Berlingieri et al. (2026)) shows that firms of different sizes choose different strategies of innovation. With uncertainty and learning integrated to the quality-ladder framework, my model allows innovation to depend on firms' history of patenting experiences, which proposes a new source of heterogeneity and allows innovation intensity to differ for firms that are of the same size but experienced different histories of signal realizations.

Second, this paper contributes to the study of direction and path of innovation. The quality ladder models typically assume that creative destruction is undirected. The expanding variety literature (Acemoglu (2002), Acemoglu et al. (2012)) discusses the conditions under which innovation is directed towards binary features of technologies (skill-based vs unskill-biased, clean vs dirty, ...). In this class of model, path dependence of innovation is generated by assuming the accumulated knowledge base increases the productivity of future innovation in a field. Similarly, Redding (2002) proposes that technology-specific secondary knowledge transfers imperfectly to the next fundamental technology, so accumulating knowledge around

the incumbent technology reduces the return to searching for a new technological trajectory. My model proposes an alternative channel of path dependence through learning. In my model, path dependence is conditional upon the history of accumulated signals: firms will not spend much of their R&D resources on fields that accumulated many low-signal patents.

Third, this paper contributes to a small literature on the relationship between innovation, uncertainty and learning. [Lei and Shi \(2024\)](#) show that Knightian uncertainty discourages entrepreneurial entry, particularly among less wealthy households. [Jo and Kim \(2025\)](#) study delays in learning other firms' frontier technologies, which induce incumbents to use own-innovation strategically to deter creative destruction. My mechanism is distinct from theirs: firms use noisy market feedback from their own patents to update Bayesian beliefs about innovation returns across technologically related fields.

Lastly, this paper is related to the learning literature. The key modeling device of the paper is Gaussian processes, which are widely used in machine learning ([Rasmussen and Williams \(2006\)](#)). In behavioral economics, [Ilut et al. \(2020\)](#), [Ilut and Valchev \(2022\)](#) and [Ilut and Valchev \(2024\)](#) adopt Gaussian processes to model uncertainty over an unknown functional relationship (such as the value function and policy function from a dynamic programming problem). Gaussian processes are also used in cognitive psychology to capture associative learning and exploration-exploitation tradeoff ([Plate, Ham, and Jenkins \(2023\)](#)). In the IO literature, [Hodgson and Lewis \(2025\)](#) find that the sequential search behavior of consumers is consistent with belief updating on Gaussian process priors. My paper imports the methodology to the innovation literature.

Layout The rest of the paper proceeds as follows. Section 2 presents a simplified, illustrative model of innovation with associative learning and sketches the key predictions of the model. Section 3 presents the empirical analysis that tests the model predictions. Section 4 specifies the details of the full quantitative model and calibration, and studies the effect of sentiment shock to the growth path. Section 5 concludes.

2 An Illustrative Example

We begin with a simplified firm innovation problem that illustrates the learning mechanism. Let $x \in X$ index technological fields. Let n_x denote the number of patents the firm holds in

field x , and let $V(n_x, x)$ denote the value of those patents. For simplicity, suppose that

$$V(n_x, x) = \lambda n_x, \quad (1)$$

where $\lambda > 0$ is the common step size of a patent. The value of an additional patent in field x is therefore

$$\Delta V(x) = V(n_x + 1, x) - V(n_x, x) = \lambda, \quad (2)$$

which does not depend on x .⁴ Hence, innovation in every field yields the same payoff. The firm chooses a nonnegative innovation intensity z at convex cost

$$c(z) = \frac{1}{2}z^2. \quad (3)$$

I treat z as an arrival intensity. If it is instead interpreted as a success probability, the expressions below describe the interior solution and z must be capped at one.

Following the quality-ladder literature (e.g. [Akcigit and Kerr \(2018\)](#)), a firm may hold patents in multiple fields. Let $\mathcal{X}_t \subset X$ denote its portfolio at the beginning of period t . The firm chooses an internal innovation intensity separately for every field $x \in \mathcal{X}_t$ and selects one previously unexplored field for external innovation. The maximized expected net payoff from an internal field x is

$$W(x) = \max_{z \geq 0} \{-c(z) + z\mathbb{E}[\Delta V(x)]\}. \quad (4)$$

The firm chooses the external field according to

$$x_t^* \in \arg \max_{x' \in X \setminus \mathcal{X}_t} \left\{ \max_{z \geq 0} [-c(z) + z\mathbb{E}[\Delta V(x')]] \right\}. \quad (5)$$

If the firm knows the primitives, then $\mathbb{E}[\Delta V(x)] = \lambda$ for every x . The interior solution is $z(x) = \lambda$ and the maximized net payoff is $W(x) = \lambda^2/2$. The firm therefore chooses the same intensity in every internal field and is indifferent among all external fields; I assume uniform random tie-breaking. This is the canonical undirected quality-ladder benchmark [Klette and Kortum \(2004\)](#); [Acemoglu \(2009\)](#); [Akcigit and Kerr \(2018\)](#). The conclusion is unchanged if field-specific step sizes are identically distributed according to a known distribution F . For a risk-neutral firm choosing z before the step size is realized, $z(x) = \bar{\lambda}$ and $W(x) = \bar{\lambda}^2/2$, where $\bar{\lambda} = \mathbb{E}[\lambda]$ ([Berlingieri et al. \(2026\)](#)). Because costs are separable across fields, this

⁴Note that here I am assuming that internal and external innovation have the same step size λ for simplicity. In the full model in Section 4, this assumption will be relaxed.

illustrative setup compares field-specific intensities; it does not impose a fixed aggregate R&D budget.

Uncertainty, Information, and Beliefs Now I introduce information friction to the firms. The firm knows the common benchmark λ but is uncertain about the effective step size of innovation in each field. It represents the step size in field x as $\lambda \exp(\tilde{v}(x))$, where $\tilde{v}(x)$ is a latent function capturing the log deviation of the field-specific step size from the common benchmark. The true mapping in this illustrative environment is $\tilde{v}(x) = 0$ for every x , so the effective step size is homogeneous across fields, although the firm does not know that the mapping is flat.

The firm represents its beliefs about $\tilde{v}(x)$ with a Gaussian process. For any two fields x and x' , the values $\tilde{v}(x)$ and $\tilde{v}(x')$ are jointly Gaussian:

$$\begin{bmatrix} \tilde{v}(x) \\ \tilde{v}(x') \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m_0(x) \\ m_0(x') \end{bmatrix}, \begin{bmatrix} k_0(x, x) & k_0(x, x') \\ k_0(x', x) & k_0(x', x') \end{bmatrix} \right), \quad (6)$$

where $m_0(x)$ is the prior mean of $\tilde{v}(x)$,⁵ and $k_0(x, x')$ is the covariance kernel, which takes the form

$$k_0(x, x') = \sigma_0^2 \exp \left(-\frac{\|x - x'\|^2}{2\varphi} \right), \quad (7)$$

where σ_0^2 is the prior variance and $\varphi > 0$ is a parameter that controls the degree of associative learning across fields. We can see that the covariance between $\tilde{v}(x)$ and $\tilde{v}(x')$ falls with technological distance, so a signal about x changes the firm's assessment of nearby fields more than that of distant fields, and φ captures how fast the decline is. Two benchmark cases for φ are worth mentioning. When $\varphi \rightarrow 0$, the covariance between the value of patenting in two fields x and x' converges to 0. In that case, the learning is only local and does not spill over to nearby fields. When $\varphi \rightarrow \infty$, the covariance between the value of patenting in any two different fields converges to σ_0^2 , so that the news about one particular field would revise beliefs about the step sizes uniformly for all fields.

Suppose that the firm successfully patents in field x_t during period t . When the innovation is disclosed, the market response provides a noisy signal about its log value:

$$s_t(x_t) = \log \Delta V(x_t) + \varepsilon_t = \log \lambda + \tilde{v}(x_t) + \varepsilon_t, \quad (8)$$

⁵Here, I assume that $m_0(x) = 0$ for simplicity.

where $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$. After removing the known common component, the normalized signal is

$$y_t(x_t) \equiv s_t(x_t) - \log \lambda = \tilde{v}(x_t) + \varepsilon_t. \quad (9)$$

Under the true homogeneous mapping $\tilde{v}(x) = 0$, the normalized signal equals the transitory market noise ε_t . The firm, however, cannot distinguish the latent field component from noise and uses $y_t(x_t)$ to update its beliefs. Gaussian conditioning gives the posterior mean and covariance

$$m_{t+1}(x) = m_t(x) + \frac{k_t(x, x_t)}{k_t(x_t, x_t) + \sigma_\varepsilon^2} [y_t(x_t) - m_t(x_t)], \quad (10)$$

$$k_{t+1}(x, x') = k_t(x, x') - \frac{k_t(x, x_t)k_t(x_t, x')}{k_t(x_t, x_t) + \sigma_\varepsilon^2}. \quad (11)$$

Here $k_t(x_t, x_t)$ is the firm's conditional variance of $\tilde{v}(x_t)$ before observing the period- t signal, and $k_t(x_t, x_t) + \sigma_\varepsilon^2$ is the conditional variance of that signal.

Figure 1 illustrates the updating mechanism on a one-dimensional technological space. The technological space is represented as a real number on the x-axis. The first signal, $y = 2.5$ at field location $x = 1$, exceeds the prior mean and therefore raises the posterior mean of the latent component $\tilde{v}(x)$ around $x = 1$. Because the Kalman gain in (10) is proportional to $k_t(x, x_t)$, the revision is strongest at the observed field and decays with technological distance. The subjective uncertainty for locations around $x = 1$ is also reduced by the revealed signal, as shown by the narrowing uncertainty band around $x = 1$ in the figure. Similarly, the second signal, $y = 0.4$ at location $x = 3$, is below the pre-signal mean at that location and therefore lowers beliefs locally around $x = 3$. In both cases, the posterior mean and uncertainty at distant fields are hardly affected because their covariance with the observed field is small.

Implications for the Firm's Innovation Path Conditional on the firm's information at time t , the expected prize in field x is

$$\mathbb{E}_t[\Delta V(x)] = \lambda \mathbb{E}_t[\exp(\tilde{v}(x))] = \lambda \exp \left(m_t(x) + \frac{1}{2} k_t(x, x) \right). \quad (12)$$

With $c(z) = z^2/2$, the firm solves $\max_z z \mathbb{E}_t[\Delta V(x)] - c(z)$, and hence the interior optimum $z_t^*(x)$ and maximized net payoff $W_t(x)$ are

$$z_t^*(x) = \mathbb{E}_t[\Delta V(x)] = \lambda \exp \left(m_t(x) + \frac{1}{2} k_t(x, x) \right), \quad (13)$$

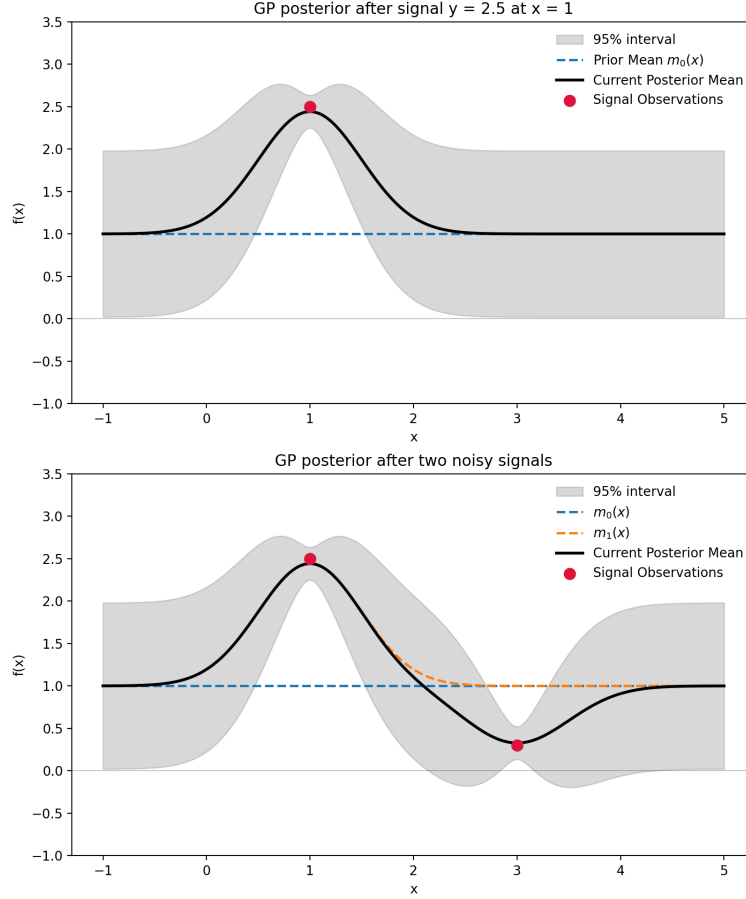


Figure 1: Illustration of Gaussian-process learning. For visual clarity, this figure uses a constant prior mean $m_0(x) = 1$ rather than the zero-centered analytical normalization; changing this level does not alter the local updating mechanism. In the upper panel, the firm observes $y = 2.5$ at $x = 1$. Equations (10) and (11) yield the posterior mean shown by the black solid line. In the lower panel, the firm additionally observes $y = 0.4$ at $x = 3$. The shaded regions show pointwise 95% posterior credible intervals.

and

$$W_t(x) = \frac{1}{2}\lambda^2 \exp(2m_t(x) + k_t(x, x)). \quad (14)$$

Because λ is common across fields, the external-field choice is equivalently

$$x_t^* \in \arg \max_{x \in X \setminus \mathcal{X}_t} \left\{ m_t(x) + \frac{1}{2}k_t(x, x) \right\}. \quad (15)$$

A favorable surprise raises $m_t(x)$ near the observed field and, holding posterior variance fixed, raises both innovation intensity and the attractiveness of nearby external fields. An unfavorable surprise works in the opposite direction. Posterior variance also matters: observations reduce $k_t(x, x)$ most near sampled fields, while distant fields remain relatively uncertain. The lognormal payoffs create an incentive for the firm to explore distant, highly uncertain fields.

To illustrate how Gaussian-process learning shapes innovation paths, I simulate one firm for 80 periods on a 15×15 grid of 225 fields. The learning parameters are $\sigma_0 = 0.5$, $\sigma_\varepsilon = 0.15$, and $\varphi = 0.5$. In the path simulations, I set the constant prior mean to $m_0 = -\sigma_0^2/2$, so that the initial expected multiplier satisfies $\mathbb{E}_0[\exp(\tilde{v}(x))] = 1$ and the learning and no-learning firms have the same expected prize before receiving signals. The figures plot successful external filings.

Figure 2 compares a no-learning benchmark with a Gaussian-process learning firm, under the same path of positive signals in the event of a successful patent. The benchmark knows that payoffs are identical across fields, receives no informative field-level feedback, and breaks ties among unexplored fields randomly. Therefore, on the left panel, we can see that the external search path is dispersed and undirected, spreading over the whole space. The learning firm instead observes field-level signals and extrapolates them across nearby technologies with $\varphi = 0.5$. Favorable early observations then raise posterior means locally and tilt subsequent innovation toward neighboring fields. The resulting path is concentrated around the initial field.

Figure 3 isolates the role of signal histories by comparing two learning firms with the same parameters and initial location. These are conditional stress tests rather than representative random draws. Including the initial patent, every successful patent generates the normalized signal $y = 0.8$ in the left panel and $y = -0.8$ in the right panel. Repeated favorable signals raise posterior means in the sampled neighborhood and produce a concentrated external path. Repeated unfavorable signals lower local posterior means and cause the firm to search

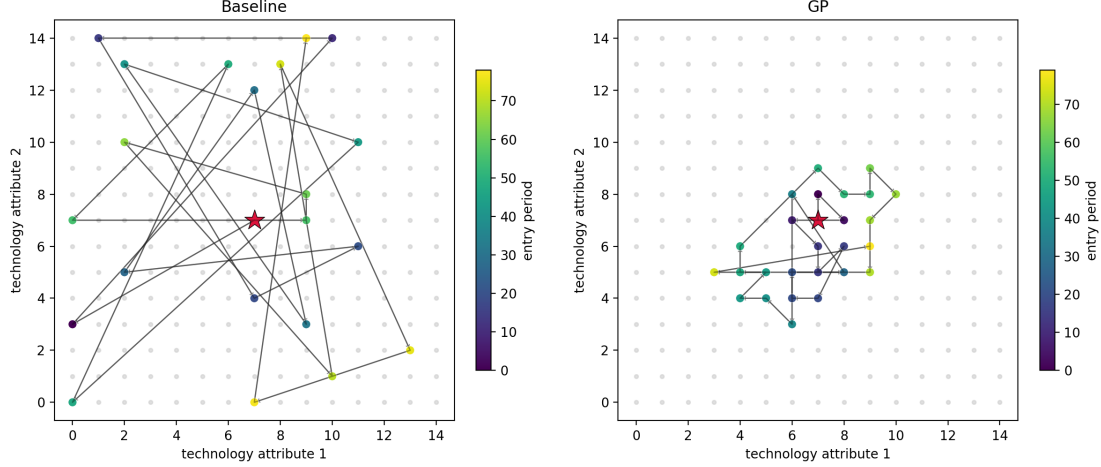


Figure 2: External innovation paths: no-learning benchmark versus Gaussian-process learning ($\varphi = 0.5$). The red star marks the initial patent, colored dots mark successful external entries, lines connect successive external entries, and colors indicate entry time.

farther away, producing larger technological steps and a more dispersed path. The contrast demonstrates conditional path dependence: identical initial technologies can lead to different innovation paths because firms experience different market signals.

Figure 4 illustrates how the extrapolation parameter φ shapes the geometry of external search. Both firms receive the same fixed normalized signal $y = 0.8$ following a successful patent. When φ is small, covariance decays rapidly with distance, so favorable signals change beliefs only at or very near the sampled field. Unexplored fields consequently remain nearly symmetric, and external entry is dispersed across the technological space. When φ is larger, the same signals raise posterior means throughout a wider neighborhood. Subsequent external innovation is therefore directed toward the region surrounding the initial field, producing a more concentrated path.

Summary In sum, the learning model has three implications. First, internal innovation intensities tilt toward fields with favorable signal histories and away from fields with unfavorable histories. Second, external innovation is more likely to occur near existing fields with high posterior evaluations and away from fields with low posterior evaluations, subject to the posterior-variance channel described above. These implications distinguish the model from the undirected quality-ladder benchmark. Third, the model generates path dependence conditional on realized signals: firms with the same initial knowledge portfolio can follow different innovation paths after receiving different feedback. This differs from canonical directed expanding-variety models [Acemoglu \(2002\)](#), in which path dependence arises

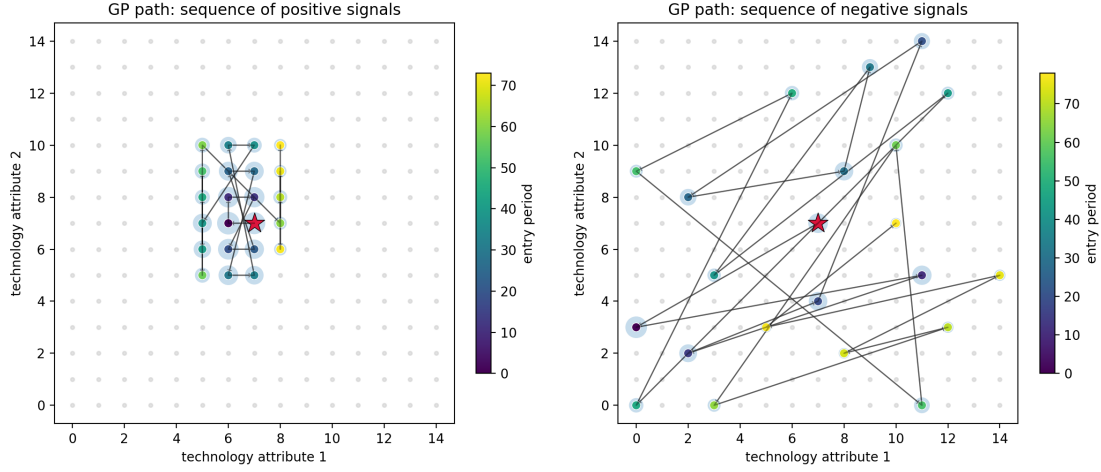


Figure 3: External innovation paths conditional on favorable and unfavorable signal histories. Both firms have $\varphi = 0.5$ and begin at the field marked by the red star. In the left panel, every successful patent, including the initial patent, generates $y = 0.8$; in the right panel, every successful patent generates $y = -0.8$. Colored dots mark successful external entries, lines connect successive entries, and colors indicate entry time.

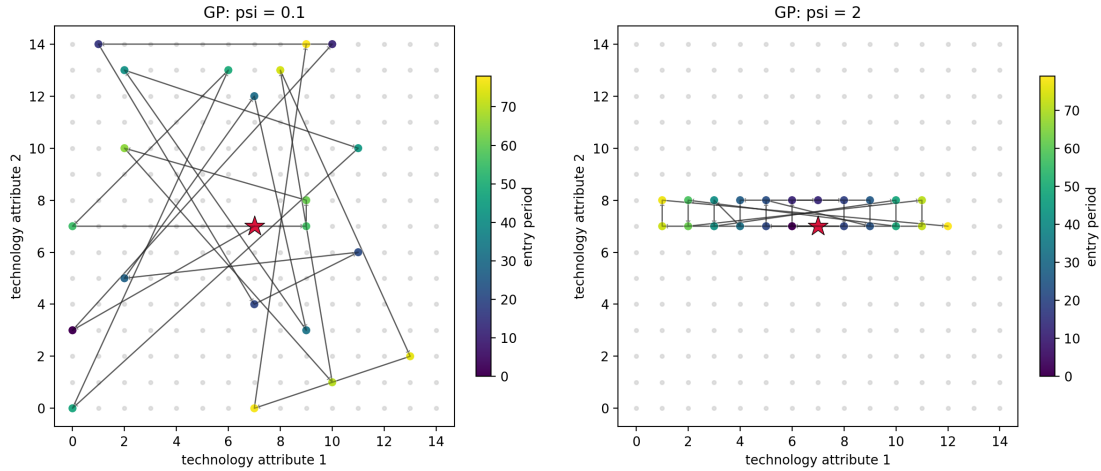


Figure 4: External innovation paths with different φ parameters. Both firms receive the same positive signal following each successful patent. The left panel sets $\varphi = 0.1$, and the right panel sets $\varphi = 2$. Colored dots mark successful external entries, lines connect successive entries, and colors indicate entry time.

from increasing returns to accumulated knowledge rather than from the history of signal realizations.

3 Empirics

In this section, I test the three predictions of associative learning from the illustrative model.

3.1 Data and Measurement

In the empirical analysis, I focus on the patenting behaviors and market responses to successful patents of U.S. public firms from 1981 to 2020. The analysis draws on two data sources. Patent data come from PatentsView, which contains the universe of patents granted by the USPTO since 1976. For each patent, the dataset provides the patent metadata, citation information and CPC classifications. The firm information, such as the public-firm identifiers, comes from the CRSP/Compustat dataset. I merge the patent- and firm-level data using the patent-firm linkage provided by [Kogan et al. \(2017\)](#).

From the illustrative model, we can see that the innovation path of firms over the technological space is crucial in distinguishing the associative-learning innovation model from other endogenous growth models in the literature. Throughout the empirical analysis, I use CPC subclasses as the empirical counterpart of “fields” X in the illustrative model. The 640 CPC subclasses $X = \{x_1, \dots, x_{640}\}$ are classified by the USPTO based on the technological features of patents and provide an appropriate level of granularity for our analysis. The key object that links theory to data is a distance measure that captures the technological similarity between two different CPC subclasses. I use patent-level citation information and construct a similarity measure based on the location of CPC subclasses in the backward citation network. Unlike the forward citation network in [Acemoglu et al. \(2016\)](#), which captures how the knowledge from a field spills over to *future* patents in other fields, the backward citation network captures how the development of patents in a field depends upon *past* knowledge from other fields, which speaks more directly to the mechanism in the illustrative model.

Specifically, for each decade $d \in \{1980\text{--}1989, 1990\text{--}1999, 2000\text{--}2009, 2010\text{--}2019\}$ and each

CPC subclass x , I first construct the share of backward citations from another field x' :

$$s_{x \rightarrow x', d} = \frac{\# \text{ of citations from field-}x \text{ patents to field-}x' \text{ patents in decade } d}{\text{Total citations from field-}x \text{ patents in decade } d} \quad (16)$$

It is the share of citations made by field- x patents that point to field- x' patents in decade d , which captures the extent to which innovation in field x depends on prior knowledge in x' . For each field x , we have a vector of backward-citation shares $\mathbf{x} \equiv (s_{x \rightarrow x_1}, s_{x \rightarrow x_2}, \dots, s_{x \rightarrow x_{640}})$, summarizing the technological dependence of field x on other fields. I then define the technological similarity between fields x and $\tilde{x}$ as the cosine similarity between vectors $\mathbf{x}$ and $\tilde{\mathbf{x}}$, as in Jaffe (1986) and Bloom et al. (2013):

$$s(x, \tilde{x}) \equiv \frac{\mathbf{x} \cdot \tilde{\mathbf{x}}}{\|\mathbf{x}\| \cdot \|\tilde{\mathbf{x}}\|} \quad (17)$$

This similarity measure is higher if two fields draw on similar knowledge. Appendix ?? illustrates the field similarity matrices for all four decades. Similar to the forward-citation matrices in Acemoglu et al. (2016), the 10-year backward-citation matrices are stable over time, and the correlations between them exceed 0.8.

The associative-learning algorithm relies on market signals generated by patenting events. The patent value in the KPSS dataset is useful for constructing a proxy for these signals. It is inferred from the firm's abnormal stock return in a three-day window around the patent grant and therefore captures the market's valuation of the patent when the grant becomes public.⁶ To isolate the patent-specific component of this valuation, I run the following regression:

$$\log \xi_i = \alpha \log(1 + C_i) + \gamma_{f(i)} + \gamma_{t(i) \times x(i)} + \varepsilon_i \quad (18)$$

where i indexes patents, ξ_i is the KPSS patent value, C_i is the number of forward citations to patent i , $\gamma_{f(i)}$ is a firm fixed effect, and $\gamma_{t(i) \times x(i)}$ is an issue-year-by-field fixed effect. The estimated residual $\hat{\varepsilon}_i$ is the component of log patent value unexplained by citations, persistent firm-level differences, and common field-year variation. $\hat{\varepsilon}_i > 0$ means that the market response to patent i was unusually favorable relative to other patents in the same field-year and other patents filed by the firm, while $\hat{\varepsilon}_i < 0$ indicates an unusually unfavorable response.

I interpret $\hat{\varepsilon}_i$ as a proxy for the patent-specific market surprise $y_t(x_t)$ in the illustrative model. Because the KPSS measure is inferred from the firm's abnormal stock return in a narrow

⁶For details on the construction of the KPSS measure, see Kogan et al. (2017).

window around the patent grant, it provides a direct measure of investors' contemporaneous response to the patent, whereas citations primarily capture technological importance. As the ChatGPT example illustrates, innovators may understand the scientific and technical merits of an innovation better than its commercial prospects. The market response can therefore provide information that is valuable to the firm. Residualizing the valuation removes standard observable and common components of patent value, paralleling the model's normalization of the observed signal by its known common component.

I then aggregate these $\hat{\varepsilon}$ signals to the firm-field-year level. Let $N_{f,x,t}$ denote the cumulative number of patents that firm f has filed in field x by year t ; this is the firm's knowledge stock used below. Because a patent's market signal becomes observable at issuance, let t_i^I denote patent i 's issue year and define the number of observed signals by

$$M_{f,x,t}^I \equiv \sum_i \mathbf{1}\{f(i) = f, x(i) = x, t_i^I \leq t\}. \quad (19)$$

For firm-field-years with $M_{f,x,t}^I > 0$, the accumulated market signal is

$$\hat{\varepsilon}_{f,x,t} \equiv \frac{1}{M_{f,x,t}^I} \sum_{i: f(i)=f, x(i)=x, t_i^I \leq t} \hat{\varepsilon}_i. \quad (20)$$

Thus, $\hat{\varepsilon}_{f,x,t}$ is the average market signal that firm f has observed by year t about field x and serves as a proxy for $m_t(x)$ in the illustrative model. This distinction keeps the filing-date knowledge stock $N_{f,x,t}$ separate from the issue-date information set used to construct signals.

3.2 Testing the Learning Mechanism

I next use these measures to test the three predictions derived from the illustrative model.

Test 1: Directional Internal Innovation The first test asks whether a firm reallocates internal innovation toward the fields in which it has received more favorable market signals. For each field x already in firm f 's portfolio at the beginning of year t , let $P_{f,x,t}$ be the share of the firm's year- t patent filings assigned to that field. The main specification is the Poisson regression

$$\mathbb{E}[P_{f,x,t} \mid \mathcal{I}_{f,t-1}] = \exp\{\alpha \log(1 + N_{f,x,t-1}) + \beta \hat{\varepsilon}_{f,x,t-1} + \delta_{f \times t} + \delta_x\} \quad (21)$$

where $N_{f,x,t-1}$ is the stock of patents that the firm had filed in field x by $t - 1$, and $\hat{\varepsilon}_{f,x,t-1}$ is the average residualized market signal from its field- x patents issued by $t - 1$. I estimate equation (21) by PPML and cluster standard errors at the firm level. The coefficient of interest is β , which measures whether the firm directs a larger share of its current filings toward an existing field with a better prior signal. Firm-year fixed effects control the firm’s total patenting in year t , along with any common change in its R&D budget or innovation opportunities. Field fixed effects absorb persistent differences in patenting intensity across CPC subclasses. The lagged patent stock is included to separate signal-driven reinforcement from the familiar knowledge-stock mechanism. Both the quality-ladder model and the Acemoglu-style directed-innovation model predict that larger existing fields attract more innovation, i.e., $\alpha > 0$, but do not predict an additional response to the realization of the market signal, i.e., $\beta = 0$. The associative learning mechanism in our illustrative model instead predicts $\beta > 0$: firms reallocate their resources internally towards fields that receive more favorable market responses.

Table 1 supports both knowledge-based and signal-dependent path dependence. The coefficient on $\log(1 + N_{f,x,t-1})$ of 1.156 confirms the knowledge accumulation channel: firms continue to innovate in fields where they have already accumulated a larger knowledge stock. More importantly, conditional on this stock, the coefficient on $\hat{\varepsilon}_{f,x,t-1}$ is 0.1178 and statistically significant at the 1% level. This means that a one-unit increase in the cumulative average signal is associated with a 12.5% increase in the expected filing share relative to the firm’s other existing fields in that year. This shows that firms actively reallocate R&D resources according to the feedback they receive from the stock market about their prior patents, which is consistent with the predictions of the associative-learning innovation model.

Test 2: Directional External Innovation The second test studies the direction of external innovation: when a firm expands beyond its existing portfolio, is it more likely to enter a candidate field that is technologically close to fields in which its prior patents received high market signals?

Unlike the internal innovation test (21), external fields have no firm-specific signal history of their own. I therefore cannot directly observe how a firm evaluates fields that it has not previously explored. Each firm has a different patent and field portfolio and history of signals, and it would be infeasible to include separate explanatory variables for each prior field because the number of explanatory variables would differ across firms with heterogeneous innovation histories. To handle this challenge, I use a flexible similarity-weighted average

Dependent Variable:	Filing share
<i>Variables</i>	
$\hat{\epsilon}_{f,x,t-1}$	0.1178*** (0.0160)
$\log(1 + N_{f,x,t-1})$	1.156*** (0.0069)
<i>Fixed-effects</i>	
Firm $\times$ year	Yes
Field	Yes
<i>Fit statistics</i>	
Observations	1,239,635
Pseudo R ²	0.41157
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Table 1: Internal innovation is directed towards high-signal fields. Each observation is at firm-field-year level. The sample covers 1981–2020. SEs are clustered by firms.

to summarize how firms use their past patenting experiences in evaluating an external field. Specifically, for every field x' outside firm f 's portfolio at $t - 1$, I define firm f 's similarity-weighted evaluation of field x' as

$$S_{f,t-1}(x'; \phi) \equiv \frac{\sum_{x \in \mathcal{P}_{f,t-1}} s(x', x)^\phi \hat{\epsilon}_{f,x,t-1}}{\sum_{x \in \mathcal{P}_{f,t-1}} s(x', x)^\phi}, \quad (22)$$

where $\mathcal{P}_{f,t-1}$ is the set of fields in the firm's portfolio before the entry decision. The logic of this construction is similar to the experience-weighting function in [Malmendier and Nagel \(2011\)](#) and [Malmendier and Nagel \(2016\)](#). Those studies weight an individual's macro-financial experience by its recency; here, the weighting dimension is technological proximity between the firm's prior experience and a candidate external field. This similarity-weighted value can be regarded as the data counterpart of $m_t(x')$ for the external field x' : the signals accumulated from an internal field x receive a higher weight in the evaluation of external field x' if the two fields are technologically similar to each other. The curvature parameter ϕ , which will be jointly estimated from the data, controls the strength of this extrapolation and plays a similar role as the parameter φ in the illustrative model. When $\phi = 0$, all technologically connected portfolio fields receive equal weight, so that the evaluations of external fields are not biased towards technologically similar internal fields. When $\phi > 0$, experience in fields more similar to x' receives greater weight. Thus, a positive estimate

of ϕ indicates that firms extrapolate prior experience locally in technological space as in the illustrative associative-learning model. Taken together, $S_{f,t-1}(x'; \phi)$ is high when the portfolio fields most similar to x' have favorable accumulated signals. Favorable experiences in nearby fields therefore contribute most strongly to the firm's evaluation of the candidate field.

Similarly, I construct a similarity-based knowledge base for the candidate external field x' as the similarity-weighted average of patent counts in the firm's existing fields:

$$N_{f,t-1}(x'; \phi) \equiv \frac{\sum_{x \in \mathcal{P}_{f,t-1}} s(x', x)^\phi \log(1 + N_{f,x,t-1})}{\sum_{x \in \mathcal{P}_{f,t-1}} s(x', x)^\phi} \quad (23)$$

which captures how much the firm knows about x' based on the knowledge accumulated in the firm's internal fields.

I next specify the test for directed external innovation. For each firm-year with at least one external-field entry, the choice set contains all CPC subclasses that are not yet in the firm's portfolio. Let $\text{Enter}_{f,x',t}$ indicate whether firm f enters external field x' in year t . I consider a Poisson regression

$$\mathbb{E}[\text{Enter}_{f,x',t} \mid \mathcal{I}_{f,t-1}] = \exp \{ \beta S_{f,t-1}(x'; \phi) + \gamma N_{f,t-1}(x'; \phi) + \delta_{f \times t} + \delta_{x'} \} \quad (24)$$

and simultaneously estimate coefficients β, γ and the weighting parameter ϕ by maximizing the likelihood. The coefficients β and γ capture the sensitivity of entry decision into a new field x' to experienced market signals and accumulated knowledge. The canonical quality ladder model predicts that β and γ should both be zero since the external search is undirected. Acemoglu's directed innovation models would predict $\gamma > 0$, i.e. firms continue to innovate in fields associated with larger knowledge stocks because of increasing returns to innovation. Neither model speaks about the weighting parameter ϕ . Under associative learning, we should expect that $\beta > 0$ and $\phi > 0$: the external innovation is directed towards fields with high similarity-weighted signals after controlling for the firm's existing knowledge, and greater weight is placed on the fields that are more similar to the candidate field x' .

Table 2 supports the predictions of the associative learning model. In Column (1), the coefficient on the similarity-based signals is positive and significant at the 1% level, after controlling for the knowledge base of the firm. To interpret the magnitude of the coefficients, for each firm in each year, I compute the difference between the 90th and 10th percentiles for the similarity-weighted signals, take the median of the gap across firm-year observations, and

multiply by the raw coefficients in Table 2. Within a firm in a given year, a candidate field at the 90th percentile of the similarity-weighted signal is about 7.7% more likely to be entered than one at the 10th percentile. The comparison indicates that U.S. public firms direct their external R&D towards fields that receive higher market signals. The estimated weighting parameter $\phi = 0.482$ implies that market responses and knowledge from technological fields that are more similar to the candidate external field receive greater weight in the evaluation of the candidate field.

Column (2) of Table 2 investigates whether the dependence of entry probabilities on experienced signals comes from entering fields with favorable similarity-weighted signals or from avoiding fields with unfavorable similarity-weighted signals. I estimate the following slope-shift version of regression (24):

$$\begin{aligned} \mathbb{E}[\text{Enter}_{f,x',t} \mid \mathcal{I}_{f,t-1}] = \exp \Big\{ & \\ & \beta^- S_{f,t-1}(x'; \phi) + \Delta\beta S_{f,t-1}(x'; \phi) D_{f,t-1}^+(x'; \phi) \\ & + \gamma N_{f,t-1}(x'; \phi) + \delta_{f \times t} + \delta_{x'} \Big\}. \end{aligned} \quad (25)$$

where $D_{f,t-1}^+(x'; \phi) \equiv \mathbf{1}\{S_{f,t-1}(x'; \phi) > 0\}$, β^- is the slope for unfavorable signals, and $\Delta\beta \equiv \beta^+ - \beta^-$ is the change in the slope when the similarity-weighted signal is positive. Thus, the slope for favorable signals is $\beta^- + \Delta\beta = \beta^+$, and the coefficient on the interaction directly tests the null hypothesis $H_0 : \beta^+ = \beta^-$. Column (2) reports an unfavorable-signal slope of $\hat{\beta}^- = 0.6432$ (SE = 0.1352), which is positive and statistically significant at the 1% level. The estimated slope difference is $\hat{\Delta\beta} = -0.5807$ (SE = 0.2212), and the null of equal slopes is rejected ($p = 0.0086$). The implied favorable-signal slope is $\hat{\beta}^+ = 0.0625$ and is not statistically different from zero. Hence, the positive pooled association between experienced signals and entry is concentrated on the unfavorable-signal side: firms disproportionately avoid candidate fields associated with more unfavorable prior market signals.

Test 3: Unfavorable Signals Push External Innovation Farther Away Test 2 showed that firms are more likely to enter candidate fields associated with favorable similarity-weighted signals. I now examine a complementary implication concerning the distance between a realized external entry and the firm's prior technological portfolio. As I have established in the illustrative model, under associative learning, a favorable signal in an existing field raises the perceived value of nearby fields and hence reinforces the external search towards these nearby fields, whereas an unfavorable signal reduces their perceived value and

Dependent Variable:	Entry	
Model:	Baseline (1)	Slope Difference (2)
Experienced Signals β	0.4268*** (0.1053)	
Knowledge base γ	2.882*** (0.0878)	2.864*** (0.0879)
Unfavorable-Signal Slope β^-		0.6432*** (0.1352)
Slope Difference $\Delta\beta = \beta^+ - \beta^-$		-0.5807*** (0.2212)
Weighing Parameter ϕ	0.482*** (0.034)	0.482 [fixed]
<i>Fixed-effects</i>		
Firm-year (choice occasion)	Yes	Yes
Candidate field	Yes	Yes
<i>Fit statistics</i>		
Observations	9,763,345	9,763,345
Squared Correlation	0.01727	0.01727
Pseudo R ²	0.17280	0.17285
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Table 2: External innovation is directed towards high-signal fields, and higher weights are placed on more similar fields. The results in Column (1) are obtained by maximum likelihood estimation. The sample period is 1990–2020. Each observation is a candidate external field for a firm in a year. The sample contains 16,742 firm-years with at least one field entry and 38,248 entry events. Column (2) keeps the weighting parameter ϕ fixed at the estimate in Column (1) and reports the slope-difference specification in regression (25); $\Delta\beta$ directly tests whether the favorable- and unfavorable-signal slopes are equal. SEs are clustered by firm for parameters other than ϕ . The SE for ϕ is computed using a firm-block bootstrap.

may redirect external search toward more distant parts of the technological space. Thus, conditional on the firm’s accumulated knowledge, the distance between an external entry and an existing field should be negatively related to the signals previously experienced in that field.

For each external field x' that firm f enters in year t , I pair the entry with every field x in the firm’s pre-entry portfolio $\mathcal{P}_{f,t-1}$. The outcome is the angular technological distance

$$d(x', x) = \arccos(s(x', x)),$$

where similarity is measured using the predetermined citation network: the network constructed in decade d is used for entry decisions in decade $d + 1$. I retain all entry-event-by-existing-field pairs rather than selecting fields according to their realized distance from the entry. I estimate

$$d(x', x) = \beta \hat{\varepsilon}_{f,x,t-1} + \gamma \log(1 + N_{f,x,t-1}) + \delta_{f \times t} + \delta_{x'} + \delta_x + u_{f,x,x',t}, \quad (26)$$

where $\delta_{f \times t}$ denotes firm-year fixed effects, $\delta_{x'}$ entered-field fixed effects, and δ_x existing-field fixed effects. Firm-year fixed effects absorb factors common to all of a firm’s external entries in a year, including its overall resources and propensity to explore. Entered- and existing-field fixed effects absorb persistent differences in the centrality and average distance of fields in the technological network. The coefficient β is therefore identified from differences in signals across the fields in a firm’s existing portfolio, conditional on their accumulated patent stocks. Associative learning predicts $\beta < 0$: an external field should land closer to an existing field for which the firm has experienced a more favorable signal.

Column (1) of Table 3 supports this prediction. The coefficient on the accumulated signal is -0.0005 and statistically significant at the 1% level. In terms of magnitude, a one-standard-deviation increase in the signal reduces technological distances by approximately 2 percentile points on average in the firm’s own distribution of technological distances.

The pooled coefficient in Column (1) can result from two sources: unfavorable signals can induce firms to leave a poorly evaluated technological neighborhood, and favorable signals may generate an equally strong movement toward an already attractive neighborhood. I decompose the pooled coefficient using the slope-shift specification

$$\begin{aligned} d(x', x) = & b^- \hat{\varepsilon}_{f,x,t-1} + \Delta b \hat{\varepsilon}_{f,x,t-1} D_{f,x,t-1}^+ \\ & + \gamma \log(1 + N_{f,x,t-1}) + \delta_{f \times t} + \delta_{x'} + \delta_x + u_{f,x,x',t}, \end{aligned} \quad (27)$$

where

$$D_{f,x,t-1}^+ \equiv \mathbf{1}\{\hat{\varepsilon}_{f,x,t-1} > 0\}, \quad \Delta b \equiv b^+ - b^-.$$

The coefficient b^- is the distance response to unfavorable signals, while $b^- + \Delta b = b^+$ is the response to favorable signals. The interaction coefficient Δb therefore directly tests whether the two slopes are equal.

Column (2) shows that the negative relationship is concentrated on the unfavorable-signal side. The estimated unfavorable-signal slope is $\hat{b}^- = -0.0011$ and is statistically significant at the 1% level. Within this region, a one-standard-deviation deterioration in the signal increases technological distance by approximately 4.6 percentile points. This confirms that firms shift their external search paths away from negative-signal fields. Interestingly, the data show limited support for a response on the favorable-signal side. The estimated slope difference is $\widehat{\Delta b} = 0.0012$ and is statistically significant at the 1% level. The implied favorable-signal slope is only $\hat{b}^+ = 0.00004$ and is not statistically different from zero. Thus, favorable signals do not appear to pull external innovation toward existing fields.

3.3 Discussion

In sum, the empirical evidence from the patenting histories of U.S. public firms is consistent with the predictions of the illustrative model. First, firms reallocate R&D resources toward high-signal internal fields. Second, they are more likely to enter new fields that are technologically associated with high-signal internal fields. Finally, the technological distance of external entry is negatively associated with signals from existing fields: firms direct external search away from the neighborhoods of low-signal internal fields.

The first limitation concerns the interpretation of the residualized market response. Although $\hat{\varepsilon}_i$ measures patent-specific public market feedback, it is neither a direct observation of the firm's belief nor an exogenous shock to that belief. The residual may contain fundamental patent quality that was already known to the firm, investors' expectations about the firm's future innovation plans, contemporaneous firm news, and measurement error. The design therefore cannot fully distinguish firms learning from market feedback from firms acting on private information that is also reflected in market valuations. The timing restrictions, fixed effects, and controls for accumulated patent stocks reduce some of these concerns but do not eliminate them. The results should consequently be interpreted as conditional behavioral patterns consistent with associative learning, rather than as causal estimates of the effect of

Dependent Variable:	Angular distance $d(x', x)$	
Model:	Baseline (1)	Slope shift (2)
<i>Variables</i>		
Accumulated signal (β in (1), b^- in (2))	-0.0005*** (0.0001)	-0.0011*** (0.0002)
Signal slope difference $\Delta b = b^+ - b^-$		0.0012*** (0.0002)
$\log(1 + N_{f,x,t-1})$	0.0001 (0.0002)	0.0001 (0.0002)
<i>Fixed effects</i>		
Firm $\times$ year	Yes	Yes
Entered field	Yes	Yes
Existing field	Yes	Yes
<i>Fit statistics</i>		
Observations	2,166,979	2,166,979
R ²	0.18385	0.18387
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Table 3: Path dependence of external search. Each observation is an entry-event-by-existing-field pair. The sample spans 1990–2020 and contains 38,251 entry events for 3,006 firms. Distance is measured on the predetermined decade citation network. Column (1) imposes a common signal slope. Column (2) allows the slope to differ above and below the zero-signal prior; the interaction coefficient directly tests whether the favorable- and unfavorable-signal slopes are equal.

stock-market feedback on R&D allocation. Future work could improve the measurement of market feedback by constructing an independent measure of investors’ patent-specific evaluations. For instance, coverages of patenting news in financial newspapers, analyst reports, and other market commentary could be linked to individual patents using firm identifiers, event dates, and textual similarity to patent abstracts. The tone of this commentary would provide a direct measure of whether investors viewed a patent’s prospects favorably or unfavorably. Residualizing this measure with respect to firm-wide sentiment, technology-wide sentiment, and observable patent fundamentals could further isolate its patent-specific and potentially nonfundamental component. Comparing such a measure with the residualized stock-market response would help validate the current proxy and clarify whether subsequent innovation responds to information about fundamentals, investor sentiment, or both.

Another limitation of the current analysis is that technological proximity is measured using a generated backward-citation network aggregated to CPC subclasses. Because citation vectors are sparse, most pairwise cosine similarities are close to zero. This concentration does not mechanically eliminate the identifying variation, but it implies that economically meaningful comparisons may depend on small—and potentially noisy—differences in the estimated network. The field-level construction may also suppress heterogeneity within CPC subclasses and reflect citation practices, including examiner-added citations, in addition to technological relatedness. The estimates should therefore be interpreted as conditional on this particular representation of technology space. Several alternatives are worth considering. One possible alternative is to aggregate CPC subclasses into broader CPC classes to reduce the number of small cosine similarities. In practice, this aggregation does not resolve the problem. The majority of resulting cosine similarities under that construction are either 0 or 1, indicating that the granularity is insufficient for this definition. A promising next step is to construct an independent text-based geometry using natural-language-processing techniques, following the methods in [Argente et al. \(2020\)](#), [Bergeaud et al. \(2025\)](#) or [Carvalho and Freund \(2026\)](#). Future work could create embeddings for each patent using patent titles and abstracts, cluster patents in the resulting semantic space, and use these representations to trace firms’ innovation paths. For the present design, continuous patent embeddings have two merits. First, they may help with clustering the patents at a more appropriate level of granularity, potentially more granular than CPC classes and coarser than CPC subclasses, so that the distribution of cosine similarities is better-behaved. Additionally, they could be used to construct predetermined field centroids and distances using only patents observed before each entry decision, after which the external-entry tests could be repeated using the text-based measure using an independently constructed text-based geometry.

4 Quantitative Model

This section embeds the associative-learning mechanism in a canonical quality-ladder environment for a quantitative analysis. The quantitative exercise has three purposes. First, it disciplines the prior uncertainty, signal noise, and spatial reach of learning using U.S. patent data. Second, it assesses whether the calibrated model reproduces the conditional path dependence documented in Section 3, including its disappearance when innovation is undirected. Third, it allows us to study how market sentiment can shape the path of innovation.

4.1 Model Setup

In the model, time is continuous and runs forever.

Household A representative consumer supplies a unit of labor inelastically to the labor market, owns the firms in the economy and consumes $C(t)$ units of final goods in each period. His lifetime utility is $\int_0^\infty \exp(-\rho t) \log(C(t)) dt$.

Final Good Producer Final goods are produced by labor and a continuum of intermediate goods $x \in [0, 1]$, with technology

$$Y(t) = \frac{L(t)^\theta}{1-\theta} \int_0^1 q(x, t)^\theta y(x, t)^{1-\theta} dx$$

where $q(x, t)$ is the quality of intermediate good x . Here, the intermediate good index x should be regarded as “fields”. The price of final good is normalized to 1, and the input and output markets are competitive. The final good producer chooses $\{y(x, t)\}$ and $L(t)$ that maximizes the profit, given prices $\{p(x, t)\}$ and wage $w(t)$.

Intermediate Good Producers Intermediate good producers sell differentiated goods in monopolistically competitive domestic market. Each field is dominated by one firm which takes the demand function into account. The production technology for intermediate good is

$$y(x, t) = \bar{q}(t) \cdot l(x, t)$$

where $\bar{q}(t) \equiv \int_0^1 q(x, t) dx$ is the aggregate quality of the good. Solving the profit maximization problem of intermediate producers and combining it with the labor market clearing condition gives us the monopolistic profit of each intermediate good x :

$$\pi(x) = \underbrace{\frac{\theta^{\theta+1}(1-\theta)^{2-2\theta}}{\theta + (1-\theta)^2}}_{\equiv \bar{\pi}} q(x)$$

The derivations are relegated to Appendix A.1.

Innovation under Rational Expectation As in [Akcigit and Kerr \(2018\)](#), intermediate good producers are engaged in two types of innovations: internal innovation improves on the fields that the firm already owns, and external innovations enter the fields in which the firm does not currently own.

Successful internal innovation in field x improves the quality of the field by λ :

$$q(x, t + \Delta t) = (1 + \lambda)q(x, t)$$

The internal innovation arrival rate $z(x)$ is chosen at a cost

$$R_z(z(x), q(x)) = \chi z(x)^\psi q(x) \tag{28}$$

where R_z is the expenditure of R&D on internally innovating x .

Similarly, successful external innovation in field x' improves the quality of the field by η , with $\eta > \lambda$:⁷

$$q(x', t + \Delta t) = (1 + \eta)q(x', t)$$

The success probability $e(x)$ is subject to a cost function

$$R_e(e(x'), \bar{q}) = \tilde{\chi} e(x')^{\tilde{\psi}} \bar{q} \tag{29}$$

where R_e is the expenditure of R&D on externally innovating x .

Under rational expectation, the step sizes η and λ are known to the firms in the economy. For an incumbent firm that has innovated in fields $\{x_1, \dots, x_k\}$ with quality portfolio $\mathbf{q} \equiv$

⁷External innovation or creative destruction is assumed to be of higher quality than internal innovation. This assumption is standard in the literature and supported empirically by [Jo and Kim \(2025\)](#).

$\{q_1, \dots, q_k\}$, the value function is

$$\begin{aligned} rV(\mathbf{q}) - \dot{V}(\mathbf{q}) &= \max_{\{z\}, e} \sum_{i=1}^k \bar{\pi} q_i - \sum_{i=1}^k \chi z_i^\psi q_i - \tilde{\chi} e^{\tilde{\psi}} \bar{q} \\ &\quad + \sum_{i=1}^k z_i [V((1+\lambda)q_i) - V(q_i)] - \tau \sum_{i=1}^k V(q_i) \\ &\quad + e \cdot \mathbb{E}_j [V((1+\eta)q_j)] \end{aligned}$$

where τ is creative destruction rate, the probability that a product is taken by another firm.

Appendix A.1 shows that the optimal internal and external innovation intensity are given by

$$z = \left(\frac{A\lambda}{\chi^\psi} \right)^{1/(\psi-1)}, e = \left(\frac{(1+\eta)A}{\tilde{\chi}^{\tilde{\psi}}} \right)^{1/(\tilde{\psi}-1)}$$

and the value function $V(\mathbf{q})$ takes the form

$$V(\mathbf{q}) = \sum_{i=1}^k Aq_i + B_k \bar{q}$$

where A and B_k are constants that are pinned down in general equilibrium. This value function consists of two components: the first part, $\sum_{i=1}^k Aq_i$, is the value of innovating on existing fields, and the second part, $B_k \bar{q}$, is the option value from external innovation. The value of each existing field x_i can therefore be written as

$$V(q_i, x_i) = Aq_i \tag{30}$$

We can see that neither the innovation intensity z_i nor the value of field $V(q_i, x_i)$ depends on field x_i .

Besides the incumbent firms, there is an exogenously given mass of entrants M_0 . The value function of entrants is

$$rV_0 - \dot{V}_0 = \max e_0 (\mathbb{E}_j [V((1+\eta)q_j)] - V_0) - \nu e_0 \bar{q}$$

where ν is an entry cost. Free entry requires $V_0 = 0$. The balanced growth path of the rational-expectation economy is characterized in Appendix A.1.

Innovation under Associative Learning In the model, the step size of each successful internal and external patents are uniform across fields (i.e. λ and η , respectively), and therefore the optimal external and internal innovation intensity and the value of each field are independent of field characteristics apart from the quality.

Now we introduce the associative learning mechanism into this quality-ladder economy. I assume that firms know the benchmark internal and external step sizes λ and η , but are uncertain about how the effective step size varies across technological fields. They represent the effective internal step size in field x as $\lambda \exp(\tilde{v}(x))$, where $\tilde{v}(x)$ is a latent function of field x that captures how the unobserved characteristics in field x map to the step size of patenting in the field. Hence, in the event of a successful internal innovation in field x with quality q , the value gain of field x is given by

$$\Delta V(q, x) = A(1 + \lambda \exp(\tilde{v}(x)))q - Aq = A\lambda \exp(\tilde{v}(x))q$$

Under our current rational-expectation setup, the true mapping is $\tilde{v}(x) = 0$ for all x , and this is not known by the firm. Instead, the firm's evaluation of field- x innovations depends on its beliefs about the latent function $\tilde{v}(x)$. I keep track of the beliefs using the same tool as in the illustrative example: firms have Gaussian Process prior $(m_0(x), k(x, x'))$ on $\tilde{v}(x)$, where $m_0(x) = 0$ is the prior mean function of $\tilde{v}(x)$, and the kernel function is $k(x, x') = \sigma_0^2 \exp\left(-\frac{\|x-x'\|^2}{2\varphi}\right)$, with σ_0^2 being the prior uncertainty and φ being the extrapolation parameter.

Firms update their beliefs using the market signals drawn from patents. In response to each successful patenting event, the stock market generates a noisy signal about the (log) value $\log \Delta V(q, x)$, denoted as $s(q, x)$:

$$s(q, x) = \log \Delta V(q, x) + \varepsilon(q, x) = \log(A\lambda q) + \tilde{v}(x) + \varepsilon(q, x)$$

which can be normalized into a signal $y(x) \equiv s(q, x) - \log(A\lambda q) = \tilde{v}(x) + \varepsilon(x)$.

In the event of a successful external innovation, the firm experiences a quality gain $\Delta q = (1 + \eta)q$. I assume that the firm's perceived prize is $(1 + \eta)q \exp(\tilde{v}(x))$, with $\tilde{v}(x)$ being the same latent function of field x as in the internal innovation. The prize for the firm in a successful external innovation is

$$\Delta V^e(q, x) = [A(1 + \eta)] q \exp(\tilde{v}(x))$$

The stock market draws a noisy signal of ΔV^e in each external patenting event

$$s^e(q, x) = \log(A(1 + \eta)q) + \tilde{v}(x) + \varepsilon(q, x)$$

which is normalized into a signal $y^e(x) \equiv s^e(q, x) - \log(A(1 + \eta)q) = \tilde{v}(x) + \varepsilon(x)$. Firms incorporate both internal innovation signals $\{y(x)\}$ and external innovation signals $\{y^e(x)\}$ in their information set. The belief updating process follows the same recursion as in (10) and (11) in the illustrative example. I assume that the firms preserve the history of all innovation signals they accumulated, even though they may lose control of fields they own in a creative destruction environment.

Fully characterizing the value and policy function under Gaussian process learning is technically challenging because the states are history-dependent and high-dimensional. To circumvent the technical challenge, I restrict my attention on a myopic policy under Gaussian process learning, in the sense that the firm values the payoff from a successful innovation but does not internalize how the resulting signal would affect future decisions and the distribution of future signals. Throughout the associative-learning block, I take A as the equilibrium coefficient implied by the rational-expectations balanced growth path. Because associative-learning firms have measure zero, their decisions do not alter this aggregate equilibrium object. Moreover, under the myopic policy, I do not solve a separate dynamic value function for these firms: A simply maps a perceived quality improvement into its current innovation payoff. After each realized success, beliefs are updated and the firm solves the static problem again. Specifically, when determining internal innovation intensity, the firm solves

$$\max_{z_x} \sum_x z_x \mathbb{E}_t^f[\Delta V(q, x)] - \chi z_x^\psi$$

which gives us the optimal innovation intensity for an internal field x :

$$z(x) = \left(\frac{A\lambda}{\chi\psi}\right)^{\frac{1}{\psi-1}} \exp\left\{\frac{1}{\psi-1} \left[m_t(x) + \frac{1}{2}k_t(x, x)\right]\right\} \quad (31)$$

We can see that the optimal internal innovation intensity under associative learning is the product of the rational-expectation innovation intensity z and a history-dependent, belief-driven “wedge”: the associative-learning firm would insert higher innovation intensity to the fields that give the firm high realizations of market signals. Under rational expectation, since the firm knows that the step sizes are identical over the technological space, market signals from new patents do not give them additional payoff-relevant information. This is not the

case for associative-learning firms: they are agnostic about the distribution of the value of patenting over the technological space, and they rely on the signals to form beliefs about it. If associative-learning firms are able to draw signals from each field for a gigantic number of times, their beliefs would eventually converge to the truth (i.e. $\tilde{v}(x) = 0$). But due to creative destruction, the duration of each firm owning a field and the number of signals that each firm can obtain from a field are finite. As a result, the beliefs of associative-learning firms about the distribution of patenting prospects across fields are generally different from the true distribution, making the internal innovation intensity directed towards the fields with higher posterior beliefs.

For external innovation, when the firm is choosing the direction and innovation expenditure, the quality of external field is not known, and hence the expected prize for each field is $\mathbb{E}_t^f[\Delta V(x)] = A(1 + \eta)\bar{q}\mathbb{E}_t^f[\exp(\tilde{v}(x))]$. The external innovation intensity is given by

$$\max_{\tilde{x}, e} eA(1 + \eta)\bar{q} \exp \left\{ m_t(\tilde{x}) + \frac{1}{2}k_t(\tilde{x}, \tilde{x}) \right\} - \tilde{\chi}e^{\tilde{\psi}}\bar{q}$$

and therefore the optimal external innovation intensity under associative learning is given by

$$e = \left(\frac{A(1 + \eta)}{\tilde{\chi}\tilde{\psi}} \right)^{\frac{1}{\tilde{\psi}-1}} \exp \left\{ \frac{1}{\tilde{\psi}-1} \left[m_t(\tilde{x}) + \frac{1}{2}k_t(\tilde{x}, \tilde{x}) \right] \right\} \quad (32)$$

and the field choice is given by

$$x^* = \arg \max_{\tilde{x}} m_t(\tilde{x}) + \frac{1}{2}k_t(\tilde{x}, \tilde{x}) \quad (33)$$

Again, we can see that the direction and intensity of external innovation are history-dependent and hinge on the posterior belief and uncertainty of the firm. Under rational expectation, firms know that patenting in any external fields would give the firm identical step sizes, and hence the external search is undirected. Revealing market signals of internal and external fields does not change the behaviors of firms. The uncertainty about distribution of innovation prospects in external fields makes learning shape the direction of external innovation: associative-learning firms would direct their external search in the technological space towards fields with higher posterior and higher uncertainty.

Stochastic Steady State under Associative Learning Since the decision rules under associative learning are history-dependent, the idiosyncratic states of firms are high-dimensional, making it difficult to characterize the stationary distribution of firms and the

balanced growth path under associative learning. To close the model, I take the approach by [Ilut et al. \(2020\)](#): for the quantitative analysis, I restrict my attention to the stochastic steady state of an economy in which associative-learning firms are measure-zero, with the rest of the economy populated by rational-expectation firms. With this assumption, the aggregate variables, such as creative destruction rate τ and aggregate growth rate g , are given by the values under rational-expectation balanced growth path. This assumption also implies that the chance of business stealing effect (a firm takes over a field from another firm) happening between two associative-learning firms is of measure zero, so that we can focus on the field takeovers within rational-expectation firms or between associative-learning and rational-expectation firms.

4.2 Calibration

I calibrate the model using simulated method of moments. In the simulation, each period is 1/50 year, and the technological space is represented by 100 fields arranged on a 10×10 grid. I simulate a panel of 1,000 firms for 2,500 periods and compute the simulated moments discarding the first 250 periods as burn-in. Note that the simulated panel should be regarded as 1,000 independent simulations of measure-zero associative-learning firms interacting with a universe of rational-expectation firms, and therefore all the aggregate equilibrium objects, such as growth rate and creative destruction rate, come from the rational-expectation balanced growth path. Each associative-learning firm in the panel starts with one field with quality $q = 1$ and Gaussian process prior $m_0(x)$ and $k_0(x, x')$ as before. Each associative-learning firm can take over a field from a rational-expectation firm at its chosen external innovation intensity, or lose a field to a rational-expectation firm at the equilibrium creative-destruction rate τ . There is no business stealing between two associative-learning firms in the panel. If a firm loses its last field, it quits the simulation panel with its identity and posterior beliefs discarded, and the simulation slot is replaced by a new one-field entrant.

There are in total 12 parameters in the model. I separate them into two categories. The three parameters in Gaussian-process learning, namely, the prior uncertainty σ_0 , signal noise σ_ε and the degree of associative learning φ , are estimated internally and will be discussed in detail shortly. The rest of the parameters are externally calibrated. Discounted factor $\rho = 0.02$ matches an annual discount factor of 0.97. The curvature of innovation cost function ψ and $\tilde{\psi}$ are set to 2, a standard value in the literature. The labor share in final good production function θ is set to match the average profitability of the economy. The

remaining five parameters, namely the step sizes λ and η , innovation cost scales χ and $\tilde{\chi}$, and entry cost ν , are taken from Akcigit and Kerr (2018). The parameter values used in the quantitative exercise are reported in the upper panel of Table 4. Solving the resulting rational-expectations balanced growth path gives us a creative-destruction rate $\tau = 0.1071$, an aggregate growth rate $g = 0.0141$, and an interest rate $r = 0.0341$. These equilibrium objects remain fixed when I vary the learning parameters.

Parameter	Interpretation	Source	Value
θ	Labor share in final-good production	External	0.1059
ρ	Discount rate	External	0.0200
λ	Internal innovation step	External	0.0509
η	External innovation step	External	0.1124
χ	Internal R&D cost	External	0.3462
ψ	Internal R&D cost curvature	External	2.0000
$\tilde{\chi}$	External R&D cost	External	4.0665
$\tilde{\psi}$	External R&D cost curvature	External	2.0000
ν	Entry cost	External	0.8303
σ_0	Prior uncertainty	Internal	1.8873
σ_ε	Signal noise	Internal	0.7990
φ	Extrapolation parameter	Internal	0.4409

Table 4: Parameter Values in Calibration

The three parameters governing the GP learning problem are selected to match three moments from the KPSS dataset by simulated method of moments. Let $\boldsymbol{\vartheta} = (\sigma_0, \sigma_\varepsilon, \varphi)'$ denote the vector of learning parameters, let $\mathbf{m}^D$ denote the three empirical moments, and let $\mathbf{m}^S(\boldsymbol{\vartheta})$ denote their simulated counterparts. I choose the parameters to minimize

$$Q(\boldsymbol{\vartheta}) = [\mathbf{m}^S(\boldsymbol{\vartheta}) - \mathbf{m}^D]' \hat{\Omega}^{-1} [\mathbf{m}^S(\boldsymbol{\vartheta}) - \mathbf{m}^D],$$

where $\hat{\Omega}$ is the estimated covariance matrix of the empirical moments.⁸

Now I explain the targeted moments and how they help pin down the learning parameters. The three parameters affect all three moments jointly, so this interpretation describes the primary source of identifying variation rather than a one-to-one assignment of moments to parameters. The first moment is the standard deviation of the patent-level residual $\hat{\varepsilon}_p$

⁸I estimate $\hat{\Omega}$ using 300 firm-block bootstrap draws of the KPSS sample. Each draw resamples firms with replacement, preserves the complete patent history of each selected firm, and re-estimates both the patent-signal first stage and all three downstream moments. I set $\hat{\Omega}$ equal to the sample covariance matrix of the resulting bootstrap moment vectors.

constructed in Section 3. Since we used the residual as a proxy for the market signal to a new patent, the signal-dispersion moment therefore provides the most direct information about σ_ϵ . The second moment is the average within-size variance of annual patent growth. Standard models imply that firms with the same number of fields have similar innovation intensities, and hence within a given firm size (measured by the number of fields the firm owns), the annual patent growth among firms should be similar. In the learning model, firms of the same size can have different innovation intensities because they have experienced different signal histories and possess different posterior beliefs. The resulting dispersion in innovation intensity generates larger within-size dispersion in patent growth than the rational-expectation case. Conditional on the signal-noise scale disciplined by the first moment, this moment is particularly informative about σ_0 , which controls the magnitude of prior uncertainty and hence the amplitude of the belief-driven wedge in innovation intensity.

The third moment resembles the correlation between external search step size and the signal realizations in the internal field in Test 3 of Section 3. Before each first entry into a new field, I define the firm’s “core field” as the field with the largest historical patent stock and calculate that field’s share within the firm’s patent portfolio as the “core share”. I then express the distance between the entered field and the core as its percentile among all external fields available to the firm before the entry. The targeted moment is the correlation between the core share and this external-step percentile. This moment captures the difference between associative-learning and canonical quality-ladder model in their spatial properties of innovation over the technological space. Under associative learning, a higher core share results from the core field accumulating more favorable signal realizations and higher posterior mean, which directs the external search towards fields close to the core field and makes the correlation negative. Under the canonical quality-ladder model, since the search is undirected, this correlation should be zero. In the data, the correlation is indeed negative. This moment is most directly informative about φ , because φ determines the extent to which signals from a firm’s accumulated experience are informative about other fields in the technological space.

Table 5 shows that the model closely reproduces the targeted moments, and the lower panel of Table 4 tabulates the estimated parameters. We can see that the estimated market signals are informative to firms, with the own-field Kalman gain being approximately 0.84. Moreover, the estimated $\varphi = 0.4409$ indicates that the associative-learning firms make inference about nearby fields from signals and that the inference decays as the distance between fields gets larger in the technological space. It lies between the limiting cases of fully local ($\varphi = 0$) and

uniform learning ($\varphi = \infty$).

Moment	Data	Model
Signal SD	0.794	0.796
Average within-size growth variance	0.591	0.591
Core-share/external-step correlation	-0.230	-0.233

Table 5: Fit of the three-parameter SMM calibration

Untargeted validation I assess the mechanism outside the targeted moments by applying versions of the empirical internal- and external-innovation tests in Section 3 to the simulated panel. These exercises are most informative about signs and the distinction between associative learning and canonical quality-ladder models with undirected innovation. Raw coefficient magnitudes involving technological distance should not be compared literally: the model contains 100 regularly spaced fields with Euclidean distance, whereas its empirical counterpart is the 640 CPC subclasses with a backward-citation-similarity geometry.

The simulated model reproduces the positive internal-allocation response to accumulated signals. The simulated coefficient for Test 1 is 0.803, compared with 0.118 in the data. Thus the direction agrees with the empirical evidence, but the parsimonious model substantially overstates the magnitude of internal reinforcement. This is not surprising: in the quantitative model, learning is the only source of heterogeneity in innovation intensity within firm sizes, and the discrepancy between the model-implied coefficient and data indicates that there are other sources of firm heterogeneity accounting for this within-size variation.

The simulated model is also consistent with patterns of external innovations we highlighted in Section 3. In the simulated analogue of the external-distance regression in Test 3 of Section 3, the signal coefficient is -0.469 under associative learning and approximately zero under an undirected no-learning policy. The associative-learning model therefore directs external entry toward fields associated with more favorable experience. In the test of direction of external innovation in Test 2 of Section 3, the simulated coefficient on the similarity-weighted signals is 2.581 in response to a 1 standard deviation increase in signals, compared with 0.313 in the data. Under no learning, the signal coefficient is close to zero and the localization profile is essentially uninformative. The estimated weighting parameter ϕ from the simulated panel is approximately 0.126, compared with the 0.483 estimated from the data. Although these estimates are not directly comparable since the simulation and data are based on different specifications of technological space, they verify that the mechanism

of the associative-learning model is consistent with the empirical patterns I documented for external search.

Taken together, the untargeted results support the qualitative mechanism: associative learning generates internal reallocation of R&D resources towards high-signal fields and directs external innovation towards the neighbors of high-signal internal fields. They are less successful as a quantitative fit to the strength of those responses.

4.3 Counterfactual: Sentiment as a Driver of Innovation Bursts

I now use the quantitative model to run a counterfactual exercise. John Maynard Keynes used the term “animal spirits” to describe the willingness of entrepreneurs to undertake long-horizon, risky investments without knowing the returns precisely. The same force applies naturally to R&D investments: the economic value of a new technology is highly uncertain, and, as I have showed, firms need external signals, such as market responses to their patents, to inform them of which technological directions appear promising. If those responses contain an episodic sentiment component, firms may mistake favorable market valuations for information about fundamental innovation opportunities.

The associative-learning model provides a convenient framework to study the implications of sentiments for innovation. In the baseline model, a successful innovation in field x generates the normalized signal

$$y_t(x) = \tilde{v}(x) + \varepsilon_t(x)$$

During a sentiment episode, I instead let the normalized signal take the form

$$y_t(x) = \tilde{v}(x) + \Delta_s \mathbf{1}\{x \in \mathcal{T}, t \in [t_0, t_1]\} + \varepsilon_t(x), \quad (34)$$

where $\mathcal{T}$ is the set of fields affected by the sentiment and $\Delta_s > 0$ is a common optimistic shift. Firms observe the combined signal but not its decomposition. They therefore update their GP posterior as if the sentiment component were information about $\tilde{v}(x)$. Note that here I am assuming that the sentiment does not change the fundamentals of the firms and the economy; it affects the firms and the economy only by shifting the belief of associative-learning firms towards a particular field for a period of time.

In the first exercise, I introduce a wave of optimism to a connected 3×3 block of nine fields

for ten years, on top of the 1,000-firm, 50-year simulated panel.⁹ The size of sentiment shift Δ_s is $2\sigma_\varepsilon$. For comparison, I simulate another panel of firms with identical innovation and signal histories, without the sentiment episode. The beliefs of associative-learning firms in the sentiment and no-sentiment simulations are therefore identical until the sentiment shock hits the economy.

Figure 5 plots the difference in the share of patents from the shocked fields between the sentiment economy and the no-sentiment benchmark. We can see that the sentiment shock can indeed shift the direction of innovation under associative-learning. During the sentiment episode, the patent share of the shocked block increases over time. At the end of the sentiment episode, the treated block accounts for 31.29 percent of annual patents, compared with 7.76 percent in the paired no-sentiment economy, a difference of 23.53 percentage points. The shares of intermediate and distant fields drop accordingly. Moreover, the effect of sentiment shock seems persistent. The patent-share difference between sentiment and no-sentiment economy in the shocked fields remains 23.24 percentage points one year after sentiment ends, 9.90 points after five years, and 6.77 points after ten years. Twenty years after the episode, the remaining difference in the shocked fields is 1.08 percentage points, and the patent share of other fields catches up gradually.

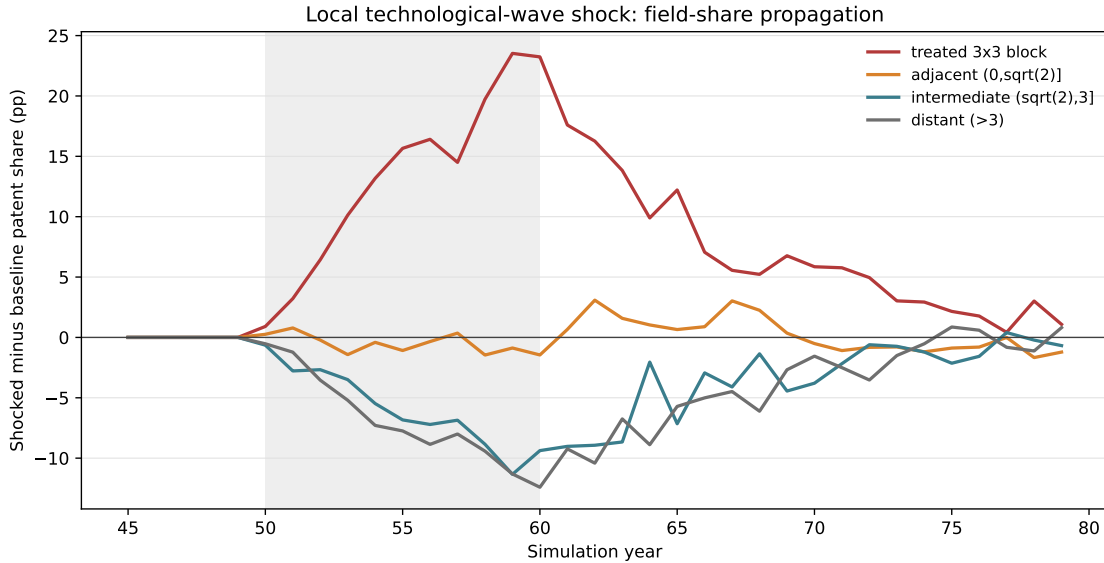


Figure 5: The Effect of a Local Sentiment Wave. The graph plots the difference of the patent share of the sentiment-shocked field between the sentiment economy and the no-sentiment benchmark. The shaded area is the sentiment episode. The history of signal realizations are identical for the two economies in the first 50 years (2,500 periods).

⁹Here, among all possible 3×3 blocks, I choose the median block in the pre-sentiment activity distribution. The shocked block accounts for 8.92 percent of learning-sector patenting before the experiment.

The result is not surprising. The wave of optimism provides the firms that have innovated in the shocked fields with persistently high realizations of signals during the sentiment episode. These firms will reallocate more R&D resources towards the sentiment fields, which increases the probability of successful patents in this fields. These new patents in turn generate more signals with persistent optimism from these fields, which further reinforces these firms' beliefs about the prospects of innovating in these fields. This feedback loop between belief and innovation is the key to generating the persistent reallocation of innovative resources in response to the sentiment shock.

Besides field-specific sentiments, the same methodology can be applied to study the implications of aggregate sentiments for innovation. After simulating a panel of 1,000 firm for 50 years (2,500 periods), I introduce the same two-standard-deviation shift to all 100 fields for 10 years, and continue the simulation for 20 more years. The upper panel of Figure 6 plots the time series of annual patents for the sentiment economy and an otherwise identical no-sentiment economy. It shows that the aggregate sentiment introduces a large and persistent patenting boom. During the sentiment episode, the learning sector produces 11,935 patents, compared with 6,894 in the paired baseline, an increase of 73.1 percent. At the end of the episode, annual patenting is 120.5 percent above baseline. Moreover, the difference persists after the sentiment fades away. After five years, the annual patents of the sentiment economy remains 44.8 percent higher than the benchmark, 24.4 percent higher after ten years, and 8.6 percent higher after twenty years.

The lower panel of Figure 6 shows that internal innovation is the main driver of the patenting boom in response to aggregate sentiment shock. Cumulative internal patenting during the sentiment episode rises by 164.5 percent, whereas external patenting rises by 26.0 percent. Even ten years after sentiment ends, internal patenting remains 53.9 percent above baseline; after twenty years it remains 20.3 percent higher. In comparison, external innovation responds more slowly and less strongly than internal innovation. The reason for this pattern is that, in our associative-learning model, the feedback loop between innovation expenditure and beliefs is much stronger for internal innovation. Since the firms do not have signals in the external fields, it's difficult for the aggregate sentiment to shift the firms' beliefs on these fields. The beliefs on the internal fields, however, keep moving upwards due to the sentiment, and the reallocation of R&D resources towards these fields generate more signals with the drift from aggregate optimism. Hence, aggregate sentiment creates the innovation boom by reinforcing the own-product innovation instead of encouraging explorations in new fields.

This pattern in associative learning is highly relevant to two salient features of stock market

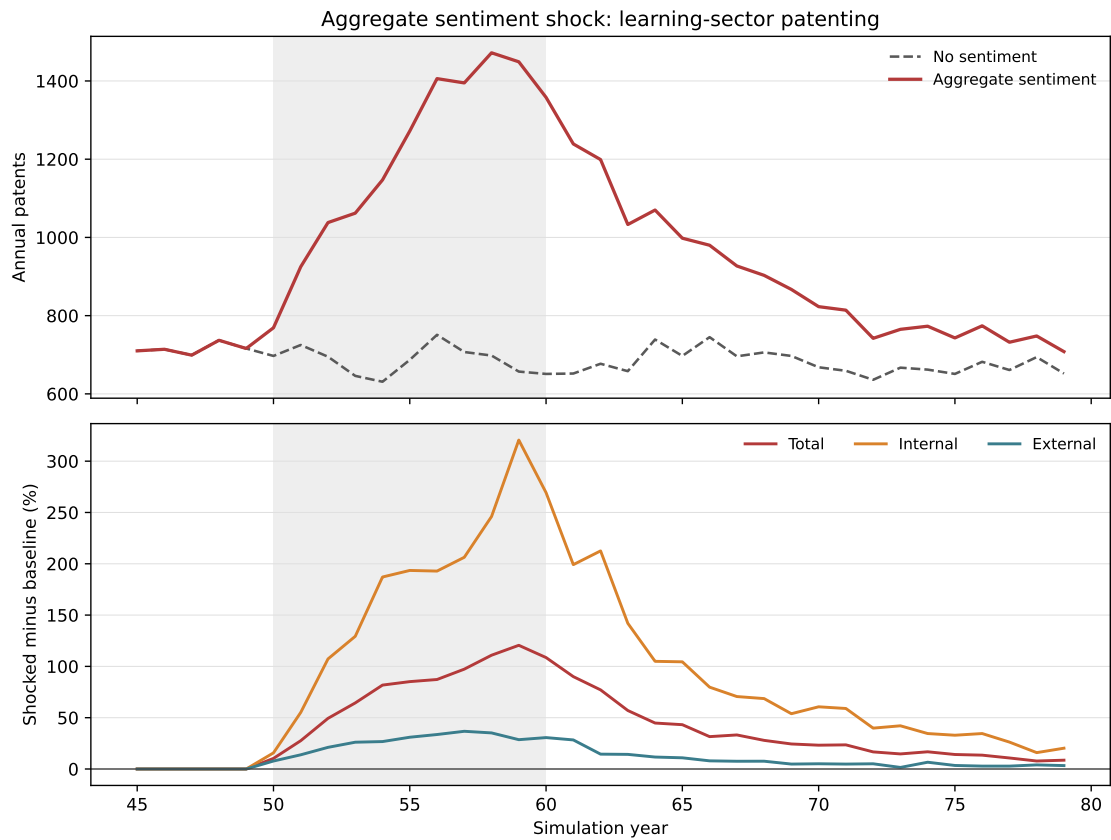


Figure 6: Aggregate Sentiment and Learning-Sector Patenting

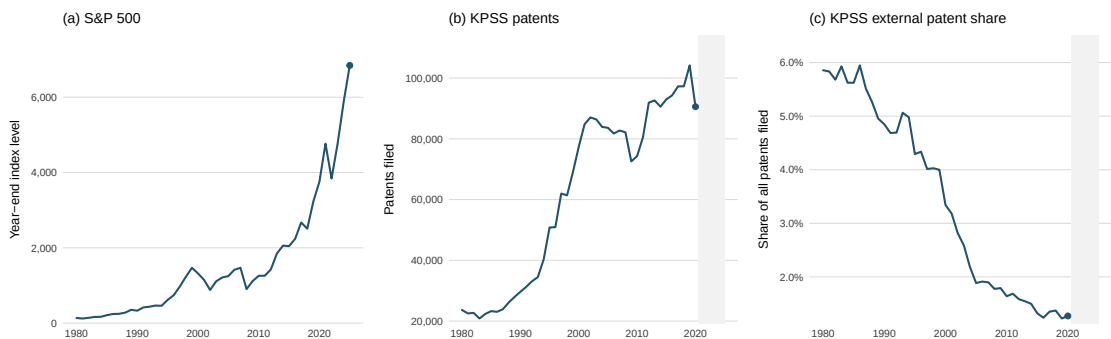


Figure 7: Recent trends of S&P500, total patent filings and share of external patents for U.S. public firms.

returns and innovation of U.S. firms in recent decades. It is well-documented that there is a downward trend of external innovation and creative destruction in the past several decades (e.g. [Garcia-Macia et al. \(2019\)](#), [Bloom \(2020\)](#), [Akcigit and Ates \(2021\)](#), [Akcigit and Ates \(2023\)](#)), while the aggregate number of patents filed by U.S. public firms has been rising persistently. In the mean time, the U.S. stock market has been persistently booming, and the firms have more resources for R&D than before. These empirical patterns are illustrated and reconfirmed in [Figure 7](#). The associative-learning mechanism provides a possible interpretation of these facts. An aggregate boom in the stock market leads the firms to update their beliefs about the values of their existing fields more and allocate more resources to internal innovation. This in turn generates more favorable signals in the internal fields. In contrast, the boom propagates to evaluations of external fields much slower because the firms need to enter the field before generating the signals. Hence the share of external innovation drops even though the aggregate patent filing is rising.

In sum, the two experiments illustrate that the associative-learning model provides a convenient and promising framework to analyze the consequences of sentiment on the path of innovation. Field-specific optimism can persistently reallocate innovation toward a popular technological region. Aggregate optimism raises the level of aggregate patenting while lowering the share of external patents.

5 Conclusion

This paper shows that when firms are uncertain about the value of patenting over the technological space, learning from experience plays a key role in the path and direction of innovation. I develop a novel quality ladder model that incorporates associative learning and allows firms to use noisy market signals to update their beliefs about innovation returns across technologically related fields. Consistent with the model, U.S. public firms subsequently direct more internal innovation toward fields receiving favorable feedback, enter new fields associated with high-signal technologies, and move away from the technological neighborhoods of fields receiving unfavorable feedback. Higher experienced signals in internal fields reduces firms' technological distances of external entry. A calibrated version of the model further shows that temporary market sentiment can produce large and persistent changes in both the level and technological composition of innovation through a feedback loop between beliefs, R&D allocation, and subsequent signals. These findings identify a source of conditional path dependence distinct from the conventional knowledge-stock mechanism: even firms with sim-

ilar technological capabilities can follow different innovation paths because they experience different histories of market feedback.

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